broader spectrum of LLMs to extend the applicability of our findings.

8 Limitation

Model Coverage This work primarily investigate the widely used LLM, Llama2, due to its prevalence in current NLG applications. However, it does not encompass other LLMs. In the future, we will extend the scope of LLMs to enhance the robustness and applicability of our results.

Human Evaluation in Data Construction Human judgment is extremely resource-intensive for hallucination judgment. The extensive time commitment and financial expenditure required are beyond the scope of this study, particularly given the large scale of the datasets. Consequently, this research did not include human evaluation in the data labeling process in § 3.2.

Comparative Performance Our method predicts the risk in advance of generation and depends solely on the query. It may lead to a trade-off in performance compared to other existing approaches that consider both the query and the response.

9 Ethical Considerations

In our experiments, we utilized datasets that are either publicly accessible or synthetically generated, thereby circumventing any potential adverse effects on individuals or communities. The datasets employed in this investigation were meticulously curated and processed to uphold the principles of privacy and confidentiality. We ensured the exclusion of any personally identifiable information, with all data undergoing anonymization before any analysis was conducted.

When contemplating the deployment of our research outcomes, we recognize the inherent risks and ethical dilemmas involved. The tendency of LLMs to produce hallucinations could disproportionately affect various demographic groups, a consequence of the inherent biases in the training datasets. We are committed to the identification and rectification of such biases to forestall the continuation of stereotypes or the inequitable treatment of any demographic.

By adhering to these ethical considerations, we aim to contribute positively to the field of NLP and ensure that advancements in understanding and mitigating hallucinations in LLMs are achieved responsibly and with consideration for the broader societal impact.

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